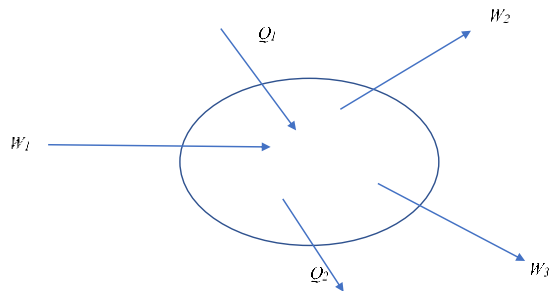


Q14



The First Law of Thermodynamics states that $\Delta U = Q + W$

where ΔU is the change in internal energy of a system, Q is the net heat inflow into the system, and W is the net work done on the system.

Hence $\Delta U = (Q_1 - Q_2) + (W_1 - W_2 - W_3)$

Q15